Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. For $A(1; 2)$ and $B(4; -2)$, calculate $\overrightarrow{AB}$.

2. Calculate the norm of $(3; -4)$.

3. Simplify $\overrightarrow{AB} + \overrightarrow{BC}$.

4. Expand $(a + b)^2$.

5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Find its cosine.

6. The midpoint of $A(-2; 1)$ and $B(4; 3)$?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Compare, on the same problem, coordinates and coordinate-free vector calculation. Justify why your method uses the data most directly.

Investigate before formalising

Construct three points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for a segment AB of 6 cm. Conjecture the locus then produce a proof that includes A and B.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.